Analysis, Question 5

$B = \frac{16 \mu_0 N I}{\sqrt{125}\, D}$. Using the fractional propagation methods from Taylor, page 61:

$$\frac{\delta B}{|B|} = \left(\left(\frac{\delta I}{|I|}\right)^2 + \left(\frac{\delta D}{|D|}\right)^2 \right)^{1/2}$$

Using the rule for inputs raised to powers (page 53):

$$\frac{\delta [B^2]}{|B^2|} = 2 \frac{\delta B}{|B|} \qquad \frac{\delta [r^2]}{|r^2|} = 2 \frac{\delta r}{|r|}$$

Then apply the principle from page 61 to $\frac{e}{m} = \frac{2V_a}{B^2 r^2}$.

$$\frac{\delta [e/m]}{|e/m|} = \left(\left(\frac{\delta [B^2]}{|B^2|}\right)^2 + \left(\frac{\delta [r^2]}{|r^2|}\right)^2 + \left(\frac{\delta V_a}{|V_a|}\right)^2 \right)^{1/2}$$

$$= \left(\left(2 \frac{\delta B}{|B|}\right)^2 + \left(2 \frac{\delta r}{|r|}\right)^2 + \left(\frac{\delta V_a}{|V_a|}\right)^2 \right)^{1/2}$$

$$= \left(4\left(\left(\frac{\delta I}{|I|}\right)^2 + \left(\frac{\delta D}{|D|}\right)^2 \right) + 4\left(\frac{\delta r}{|r|}\right)^2 + \left(\frac{\delta V_a}{|V_a|}\right)^2 \right)^{1/2}$$

$$= \left(\left(\frac{\delta V_a}{V_a}\right)^2 + 4\left(\frac{\delta I}{I}\right)^2 + 4\left(\frac{\delta r}{r}\right)^2 + 4\left(\frac{\delta D}{D}\right)^2 \right)^{1/2}$$

(noting $V_a, I, r, D > 0$.)

Additional Questions

1. Before crossing the potential, the electron has potential energy eV_a. It finishes with a kinetic energy of $\frac{1}{2} m v^2$.

$$eV_a = \frac{1}{2} m v^2$$

$$\Rightarrow 2eV_a = mv^2 \Rightarrow v = \left(\frac{2eV_a}{m}\right)^{1/2}$$

$$v = \left(\frac{2 \cdot (1.6021\ldots \cdot 10^{-19} C) \cdot (2000\ V)}{(9.1093\ldots 10^{-31})} \right)^{1/2} \approx 3 \cdot 10^7 \frac{m}{s}$$

Using the uniform circular motion constraint and the magnitude of the Lorentz force on the electron:

$$m\frac{v^{\cancel{2}}}{r} = |-e\vec{v}\times\vec{B}| = e\cancel{v}B$$

$$\Rightarrow B = \frac{mv}{er} = \frac{(9.1093\ldots\cdot 10^{-31}\ \text{kg})(3\cdot 10^{7}\ \text{m/s})}{(1.6021\ldots\cdot 10^{-19}\ \text{C})(0.35\ \text{m})}$$

$$\approx 0.0005\ \text{T} = 0.5\ \text{mT}$$

$$= 500\ \mu\text{T}$$

2 $(\gamma-1)mc^2 = eV_a$

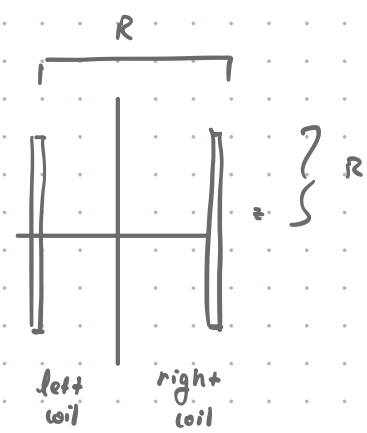
$$\gamma - 1 = \frac{eV_a}{mc^2} \Rightarrow \gamma = \frac{eV_a}{mc^2} + 1$$

For $V_a = 2000\text{V}$ and e, m, c from CODATA values:

$$\gamma = \frac{e(2000\text{V})}{mc^2} + 1 \approx 1.0039$$

This Lorentz factor isn't yet much greater than 1, so relativistic corrections are not strictly necessary.

3 Suppose two coils are placed symmetrically about the origin and parallel, with a spacing of R, their radius.



Magnetic field contributions from different contributors superpose. In a Helmholtz coil both field contributions are in the same direction, so the contributions superpose constructively.

The contribution from the left coil B_L equals the contribution from the right B_R.

Define $B_L = B_R = \vec{B}$.

$$\vec{B} = \frac{\mu_0 I R^2}{2(R^2 + R^2/4)^{3/2}}.$$

The total contribution is

$$B = 2\hat{B} = \frac{\mu_0 I R^2}{(R^2 + R^2/4)^{3/2}} = \frac{\mu_0 I R^2}{(5R^2/4)^{3/2}} = \frac{\mu_0 I R^2}{5\sqrt{5} \cdot R^3/8}$$

$$= \frac{8 \mu_0 I}{5\sqrt{5}\, R} = \frac{16 \mu_0 I}{\sqrt{125}\, D}$$

where D is the coil diameter.

With N turns in each coil, $\hat{B}$ grows by a factor of N relative to its single-loop value:

$$B = 2 \cdot N \cdot \hat{B} = \frac{16 \mu_0 I N}{\sqrt{125}\, D}$$

4 For $V_a = 2000V$ $I = 0.1238 \pm 0.0001$ A was used.

The coil diameter was 15.5 ± 0.1 cm.

$$B = \frac{16 \mu_0 (0.1238\ A)(320)}{\sqrt{125}\,(0.155\ m)} = 0.00045\ldots\ T$$

$$\delta B = B \left(\left(\frac{\delta I}{I}\right)^2 + \left(\frac{\delta D}{D}\right)^2 \right)^{1/2}$$

$$= B \cdot \left(\left(\frac{0.0001}{0.1238}\right)^2 + \left(\frac{0.001}{0.155}\right)^2 \right)^{1/2}$$

$$\approx 3 \cdot 10^{-6}\ T$$

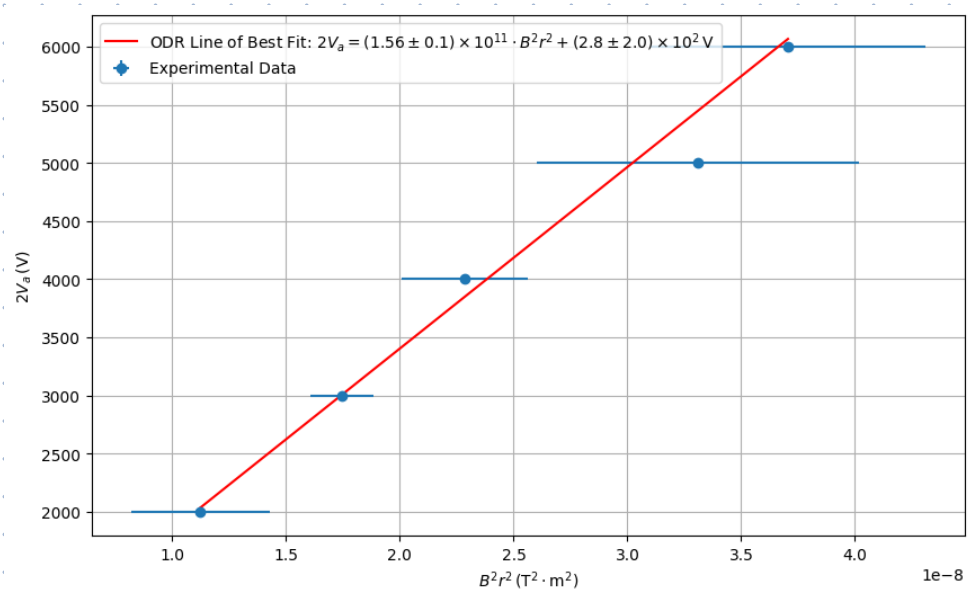
$$B = (459 \pm 3)\ \mu T$$

The value obtained from problem 1 was $500\ \mu T$, with is reasonably close to the estimated value of B for the 2000V trial.

The horizontal component of Earth's magnetic field is about $25\ \mu T$. This is not within the uncertainty computed above; Earth's field is likely to affect the experiment.

This procedure addressed this by computing y_{offset} values to correct y-coordinate readings with, obtained from a beam deflection corresponding to a 0 Helmholtz field. Subtracting this from y-coordinates aims to remove the influence of Earth's field.

This experiment also flipped the magnetic field and computed averages of y-coordinate magnitudes, reconciling for asymmetric deflection downward or upward.



$$\frac{e}{m} = \frac{2V_a}{B^2r^2} \Rightarrow 2V_a = \left(\frac{e}{m}\right) B^2r^2,$$

so $2V_a$ vs. B^2r^2 was plotted to more directly yield $\frac{e}{m}$ in the slope of the linear regression model.

This method yields $\frac{e}{m} = (1.56 \pm 0.1) \cdot 10^{11} \frac{C}{kg}$.